EHSAN SHAHINFAR

Mechanical Design Engineer | CNC Machining & Manufacturing

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PROFESSIONAL SUMMARY

Mechanical Engineer with 15+ years of experience in mechanical design, industrial equipment, CNC machining, manufacturing, plant construction, installation, and commissioning. Experienced in both engineering and hands-on manufacturing environments, with a strong background in industrial machinery, mechanical components, conveyor systems, hydraulic systems, and steel-production facilities. Currently working in Germany in CNC machining and manufacturing, with practical experience operating and programming CNC boring machines and lathes, including G-code programming and CNC control systems such as Siemens SINUMERIK 840D/810 and Heidenhain iTNC 530. Strong CAD and engineering background using Autodesk Inventor, SolidWorks, AutoCAD, AVEVA E3D/PDMS, combined with practical knowledge of manufacturing, technical drawings, tolerances, industrial standards, installation, and commissioning.

PROFESSIONAL EXPERIENCE

GETRIEBETECHNIK Dessau GmbH — Dessau-Roßlau, Germany

CNC Machining / Mechanical Engineer | 2026 – Present

- Operation and setup of CNC boring machines and CNC lathes for precision machining of mechanical components.
- Creation, modification, and optimization of G-code programs for CNC machining operations.
- Working with Siemens SINUMERIK 840D/810 and Heidenhain iTNC 530 CNC control systems.
- Reading and interpreting technical drawings, machining specifications, dimensions, and tolerances.
- Workpiece setup, tool selection, tool-offset adjustment, and machining parameter optimization.
- Performing turning, boring, drilling, and related machining operations according to technical requirements.
- Inspection and dimensional measurement of machined components to ensure required accuracy and quality.
- Troubleshooting CNC programs, tooling, and machining processes and supporting manufacturing of complex mechanical components.

SAFA FOULAD CO. — Iran

Mechanical Engineer / Head of Technical Office | Apr 2017 – 2026

KGPC Paper Production Plant — EPC Project

- Mechanical equipment design and preparation of detailed engineering drawings.
- Designed 45+ belt conveyors and 8 chain conveyors, improving equipment performance and reliability.
- Developed the plant PFD with a focus on reducing water and power consumption.
- Contributed to plant layout improvements to reduce costs and simplify operation and maintenance.
- Prepared technical project documentation, including datasheets, MRQs, MDR, and related engineering documents.
- Designed the plant dedusting system, including ducting and cyclones.

SABA Steel Company Expansion — EPC Project | In association with Danieli

- Erection, commissioning, and modification of hydraulic systems for EAF ladle preheaters.
- Erection and commissioning of bag filter and wet scrubber dedusting systems.
- Led renovation projects for Steelmaking, Thin Slab Caster, and Hot Strip Mill facilities.
- Installation and commissioning of hydraulic systems throughout the steel production line.
- Supported EAF renovation, teeming and charging crane installation, steel structure installation, and process optimization.
- Thin Slab Caster: mold and segment preparation, oscillator installation, tundish pre-heater installation, renovation, and banana-frame adjustment.
- Hot Strip Mill: descaling pump installation and commissioning and addition of the seventh mill stand.

CHARKH GARDOON CO. — Iran

Co-Founder / Mechanical Engineer | Jan 2014 – Apr 2017

Soud Gostar Power Plants Project

- Designed and constructed a special hydraulic crane for installation of power-plant motors and generators weighing approximately 40 tonnes each.
- Reduced equipment installation costs by more than 50%.
- Managed installation of approximately 1,000 tonnes of equipment and machinery.
- Assembly of hydraulic power units and cylinders.

Electro-Hydraulic Grab — Steel Production Plant Clarifier

- Designed and fabricated the mechanical mechanism and hydraulic circuit.
- Checked and inspected machined components for quality compliance.
- Designed, constructed, and assembled cable reel systems.

SAFA FOULAD CO. — Iran

Mechanical Engineer | Jan 2010 – Jan 2014

Two 30,000 Nm^3 Oxygen Plants — EPC Project | In association with Hangyang Co.

- Installation and commissioning of two 30,000 Nm^3 oxygen plants.
- Worked with international equipment suppliers including Hangyang, Atlas Copco, and Ingersoll-Rand.
- Developed the Oxygen Plant Cold Box model using PDMS.
- Head of the technical office for Cold Box erection and commissioning.
- Conducted commissioning activities for Atlas Copco air compressors.

CORE TECHNICAL SKILLS

CNC Machining & Manufacturing
CNC Turning / Lathe • CNC Boring • G-code Programming • Siemens SINUMERIK 840D/810 • Heidenhain iTNC 530 • Machine Setup & Operation • Tool Selection & Offset Correction • Machining Parameter Optimization

Mechanical Design & CAD
Autodesk Inventor • iLogic • iPart / iAssembly • SolidWorks • AutoCAD • AVEVA E3D / PDMS • Mechanical Component & Machine Design • Technical Drawing & Documentation

Industrial Equipment & Conveyors
Belt Conveyor Design • Conveyor Geometry & Power Calculations • Belt Analyst • Helix DeltaT • CEMA / DIN / ISO • Dedusting Systems • Ducting • Cyclones • Hydraulic Systems

Engineering & Commissioning
Equipment Installation & Alignment • Mechanical Assembly • Hydraulic Systems • Plant Construction & Commissioning • Quality Inspection • NDT (RT, VT, PT) • ASME / DIN / ISO / CEMA / FEM

Programming & Engineering Analysis
Python (Pandas, NumPy, scikit-learn) • MATLAB • Engineering Data Analysis • Engineering Calculations & Process Optimization

Manufacturing & Engineering
Precision Machining • Technical Drawing & Tolerance Interpretation • Manufacturing Support • Process Troubleshooting • Mechanical Problem Solving

EDUCATION

Yazd University — Yazd, Iran | M.Sc. Mechanical Engineering | Sep 2006 – Apr 2009
Chamran University of Ahwaz — Ahwaz, Iran | B.Sc. Mechanical Engineering | Sep 2001 – Apr 2006

LANGUAGES

English: Fluent — IELTS 7 German: B2 — Intermediate French: B1 — Intermediate

PROFESSIONAL CERTIFICATION

First-Grade Supervising Engineer – Building Mechanical Facilities — National Building Regulations Affairs Office, Iran